

Supplementary Information for

State-correlated reaction dynamics unveiled in full from a single product-image measurement

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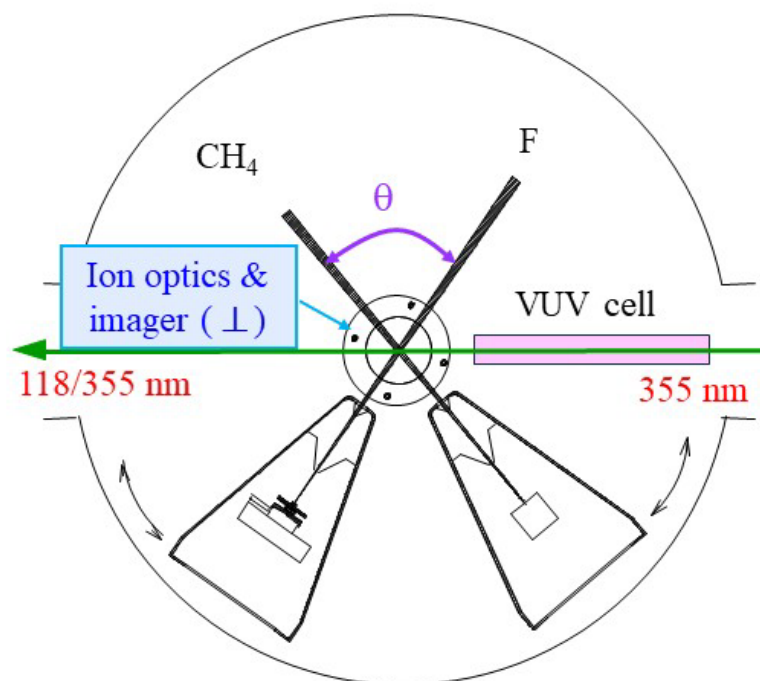
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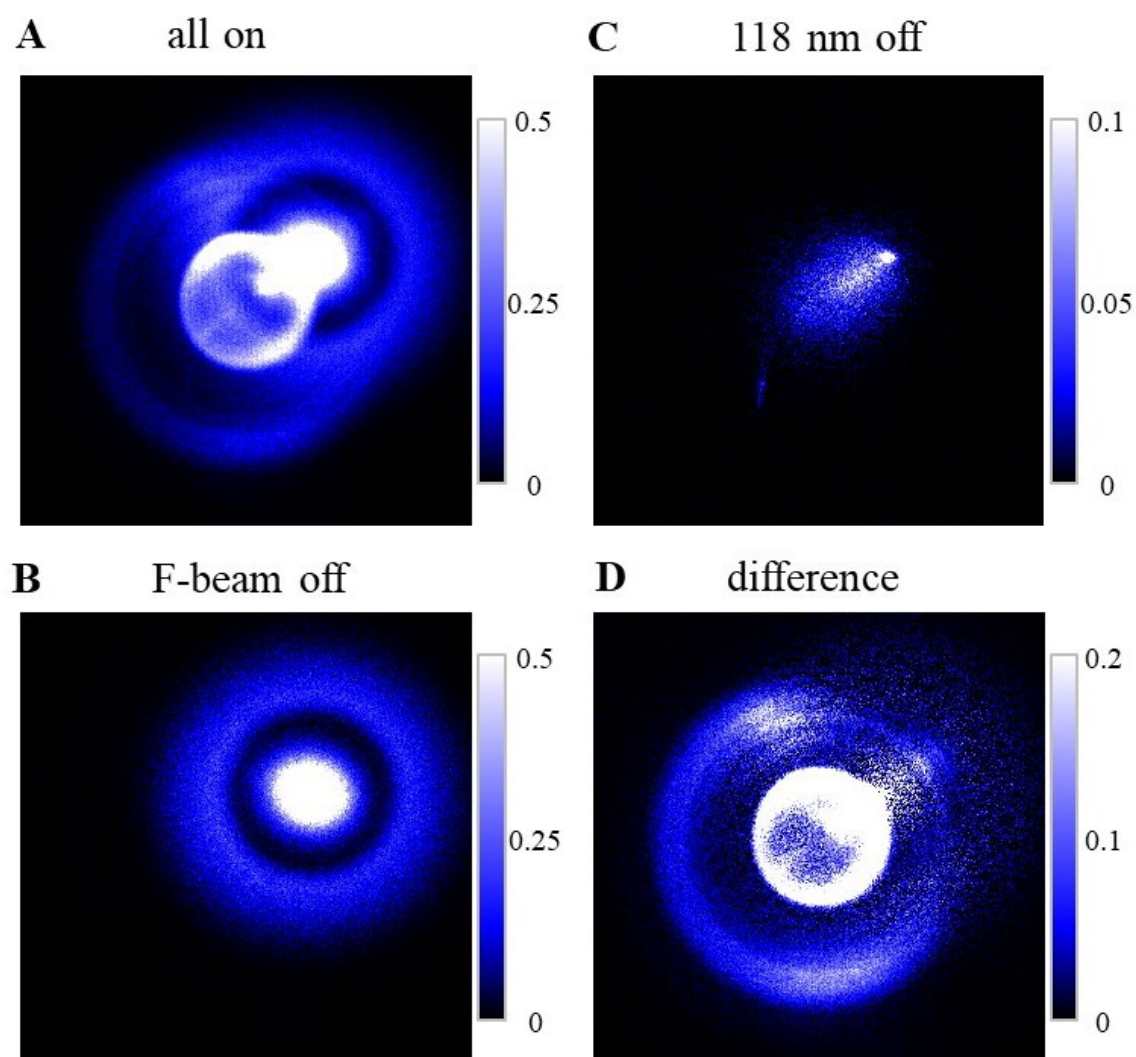
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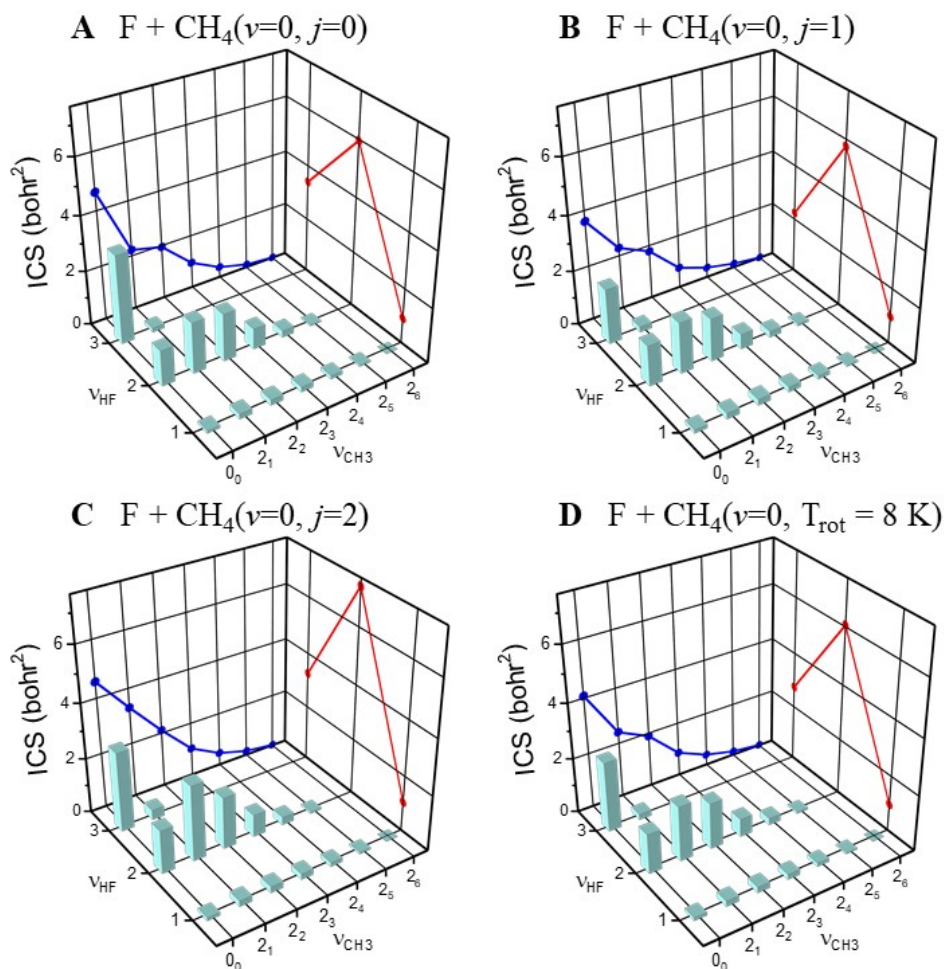
Supplementary Fig. 1–Fig. 10



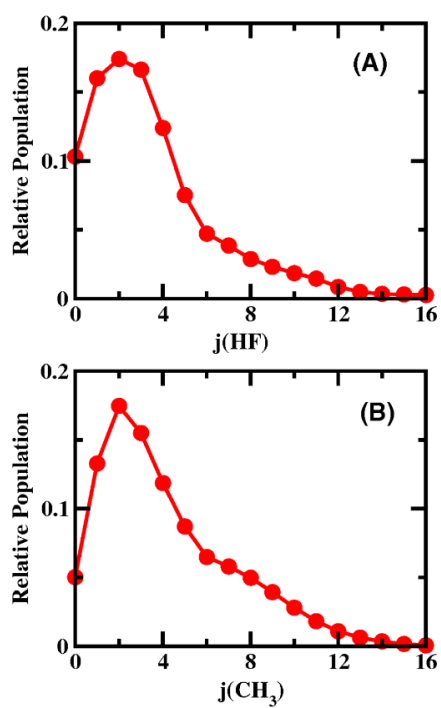
Supplementary Fig. 1 | Schematic of crossed-beam apparatus. Detailed setup of the time-sliced 3D velocity-map imager (VMI) can be found in refs. 12 (fig. 1) and 51 (extended data fig. 1).



Supplementary Fig. 2 | Raw images of the reaction products. Three raw images A, B, and C were alternatively acquired. (A) corresponded to reactive signals plus backgrounds, whereas (B) and (C) arose from the photochemical background of the reactant CH₄ at 118 nm and the discharge-generated background, respectively. By subtracting the image signals of B and C from A, the signals of the reactive event of F + CH₄ → CH₃ + HF at $E_c = 2,4 \text{ kcal mol}^{-1}$ was recovered from the background-subtracted image (D), which is presented in Fig. 1A. The color bars on the right indicate the corresponding relative intensity in linear scale. Source data are provided as a Source Data file.

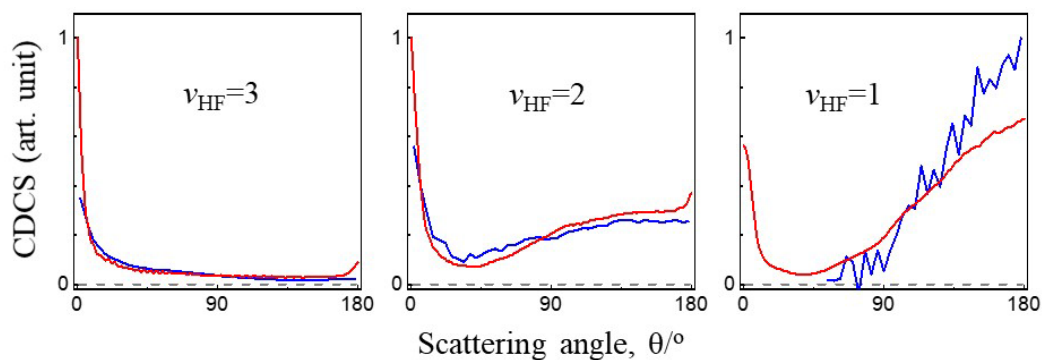


Supplementary Fig. 3 | Effects of initial rotational excitation of $\text{CH}_4(v = 0)$ reactant on the product pair-correlated distributions: (A), (B), and (C) are for the rotational states of $j = 0, 1$, and 2, respectively. (D) is for CH_4 at the rotational temperature of 8 K ($n_{j=0} : n_{j=1} : n_{j=2} = 0.31 : 0.54 : 0.15$), which is presented in Fig. 3 when compared with experiment. The most prominent difference between A and D is a significant reduction of the HF($v = 3$) branching fraction with rotational excitation of CH_4 . Since HF($v = 3$) is predominantly formed by the resonance pathway, this comparison attests a notable negative effect on the resonance contribution from rotationally excited reactants. Conceivably, when including rotationally excited CH_4 , the calculated forward peak in product angular distribution—the imprint of resonance pathway—would be smaller than those shown in Fig. 4, bringing better agreements with experiment. Source data are provided as a Source Data file.

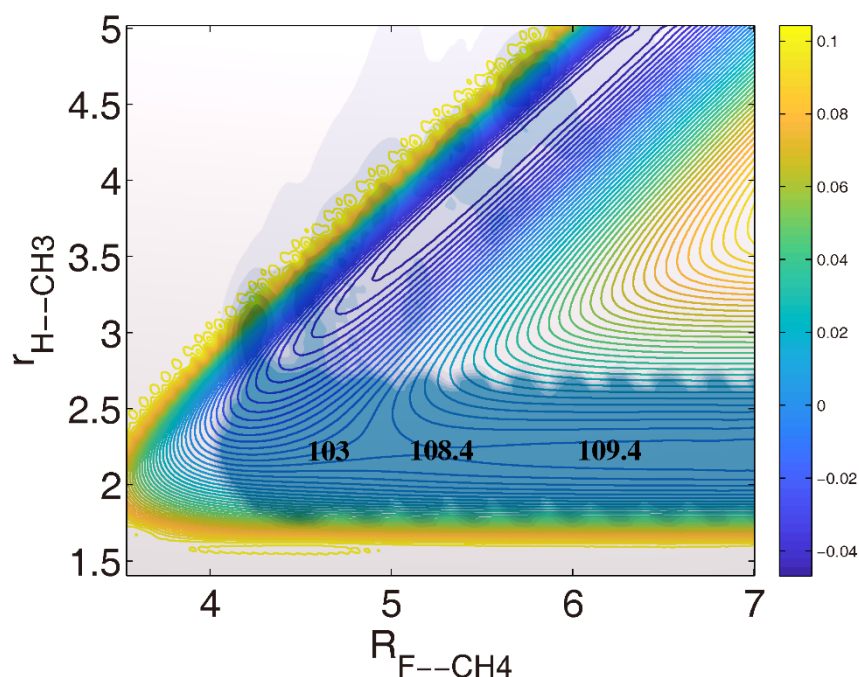


Supplementary Fig. 4 | Rotational disposal in the reaction of $F + CH_4 \rightarrow HF + CH_3$. QD rotational state distributions for (a) HF and (b) CH₃ product at $E_c = 2.4$ kcal mol⁻¹ (0.106 eV).

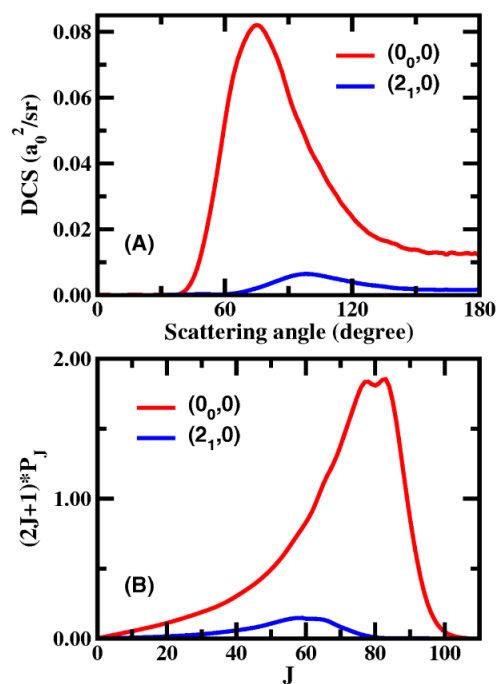
5 Source data are provided as a Source Data file.



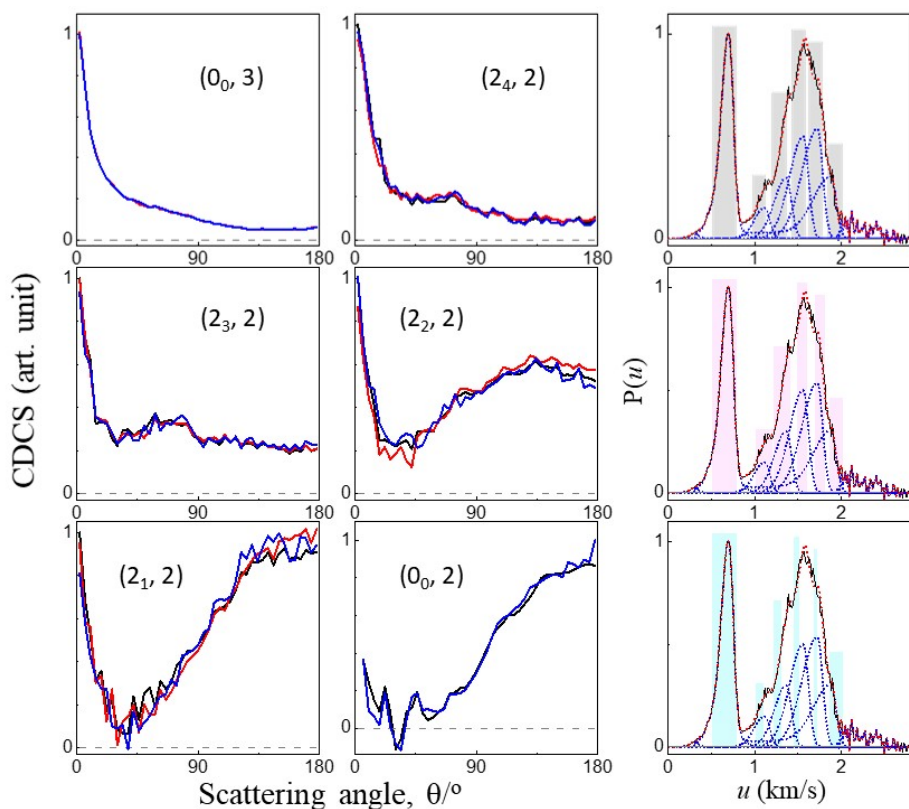
Supplementary Fig. 5 | Vibrationally state-resolved angular distributions of the HF(ν) products. A strong kinematic correlation between the scattering angle and angular momentum disposal is manifested as a systematic, recoiled-energy dependent shift from the forward-scattered HF($\nu = 3$) to mainly backward-scattered HF($\nu = 1$) state. The attributes of pair-correlated CH₃ coproducts were grouped into the corresponding HF(ν) state as in the conventional DCSs for HF(ν). The blue profile is from experiment and the red is by 6D-QD calculations. Source data are provided as a Source Data file.



Supplementary Fig. 6 | 2D representation of PES and scattering wavefunctions. The contour lines represent the potential energy (in hartree) along the two reagent Jacobi coordinates $R(\text{F}-\text{CH}_4)$ and $r(\text{H}-\text{CH}_3)$, obtained by minimizing all other degrees of freedom. The three angles marked on the diagram correspond to the umbrella-bending-angles minimized potential energy at the transition state, as well as before and after the transition state. Reactive scattering wavefunctions at the collision energy of $2.4 \text{ kcal mol}^{-1}$ are also plotted in the reagent Jacobi coordinates $R(\text{F}-\text{CH}_4)$ and $r(\text{H}-\text{CH}_3)$, with the umbrella angle maximized and other coordinates integrated. The region around the transition state is obscured by the non-reacting wave function, which is reflected back. Source data are provided as a Source Data file.



Supplementary Fig. 7 | The $\text{Cl} + \text{CH}_4 \rightarrow \text{CH}_3(v_i) + \text{HCl}(v)$ reaction at the collision energy of 0.3 eV. (A) The QD product vibrational state resolved DCS, and (B) the opacity functions. Source data are provided as a Source Data file.



Supplementary Fig. 8 | Test of the consistency of the correlated differential cross sections

(CDCSs). Displayed in each panel of two left columns are three color-coded (black, red, and blue)

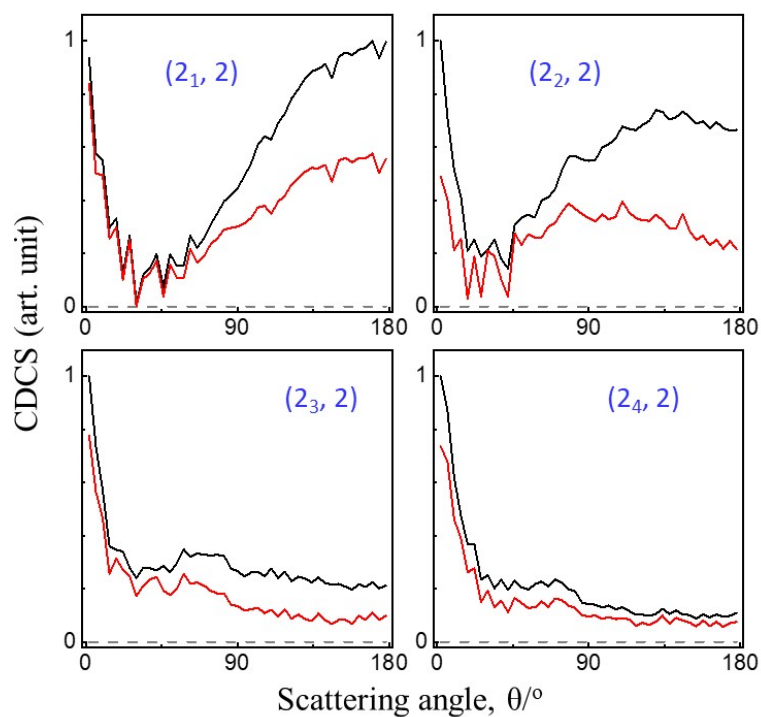
CDCSs of a denoted product pair, which are derived from distinct ranges of product speed u . The

5 u -ranges used are shown in the right panel with three corresponding, color-coded annuli (in lighter color for clarity) covering different u -ranges from broader (top, in grey) to narrow (bottom, in blue).

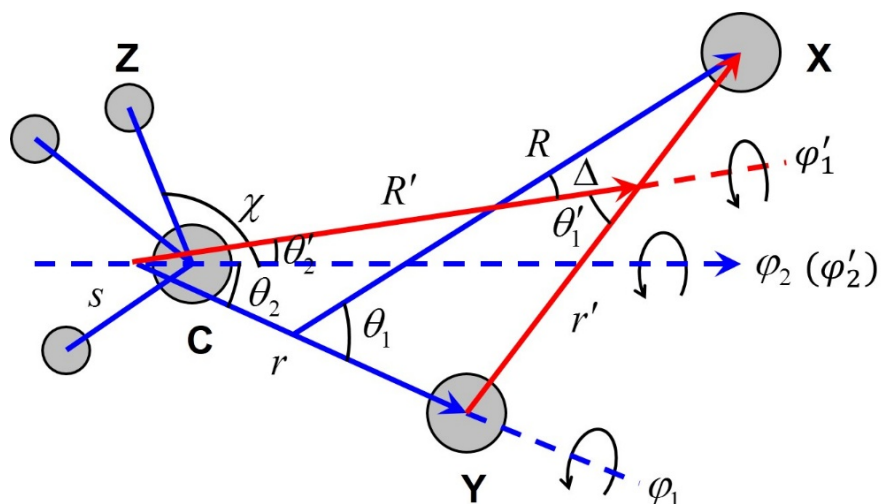
To avoid the crowdedness of the state-pair labeling, see Fig. 2 for the assignment of the dissected speed components or product pairs. As seen, the CDCSs are insensitive to the u -ranges used. For

more details, see Methods, Image analysis in the main text. Source data are provided as a Source

10 Data file.



Supplementary Fig. 9 | Effects of the overlapped contaminations on the CDCS. Depicted are the CDCSs before (black) and after (red) removing the overlapped contaminations from the proximate product pairs. The red-shaded slabs or the u -ranges in Supplementary Fig. 8 are used to analyze the CDCSs presented here. Source data are provided as a Source Data file.



Supplementary Fig. 10 | Coordinates used in the 6D-QD calculation. The reagent Jacobi coordinates $(R, r, s, \chi, \theta_1, \theta_2, \phi_1, \phi_2)$ for the X+YCZ₃ arrangement and product Jacobi coordinates $(R', r', s, \chi, \theta'_1, \theta'_2, \phi'_1, \phi'_2)$ for the XY+CZ₃ arrangement. Δ is the angle between R and R' .